DSA06 DATA HANDLING AND VISUALIZATION

Activity: Map Storytelling Challenge | Slot D

Name: U ROSHITHA
Reg. No: 192425184
R. No: 38
Story Brief: MS-D38, High-school graduates (% of adults, 1970)
Comparison variable: Per-capita income (dollars, 1974)
Focus region: South
Tool: R (ggplot2, maps)

Set-up

The set-up code given in the question was run first, without any changes.

Code:

```
library(ggplot2); library(maps)
d <- data.frame(state = state.name, zone = state.region, state.x77)
names(d) <- c("state", "zone", "pop", "income", "illit", "life", "murder", "hsgrad", "frost", "area")
d$id <- tolower(d$state)
mapdat <- merge(map_data("state"), d, by.x = "region", by.y = "id")
mapdat <- mapdat[order(mapdat$group, mapdat$order), ]
V <- d$hsgrad        # the variable you map
W <- d$income        # the comparison variable
```

Task 1: The base map

Code:

```
p1 <- ggplot(mapdat, aes(long, lat, group = group, fill = hsgrad)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_c(option = "mako", name = "High-school\ngraduates\n(% of adults)") +
  labs(title = "High-school graduates (% of adults), 1970",
       caption = "Source: US Census Bureau, R dataset state.x77") +
  theme_void(base_family = "Times New Roman", base_size = 12)
print(p1)
hi <- which.max(V); lo <- which(V == min(V))
cat("Highest state:", d$state[hi], "with", V[hi], "\n")
cat("Lowest state :", d$state[lo], "with", V[lo], "\n")
cat("Median of all 50 states =", median(V), "\n")
```

Output:

```
Highest state: Utah with 67.3
Lowest state : South Carolina with 37.8
Median of all 50 states = 53.25
```

Map:

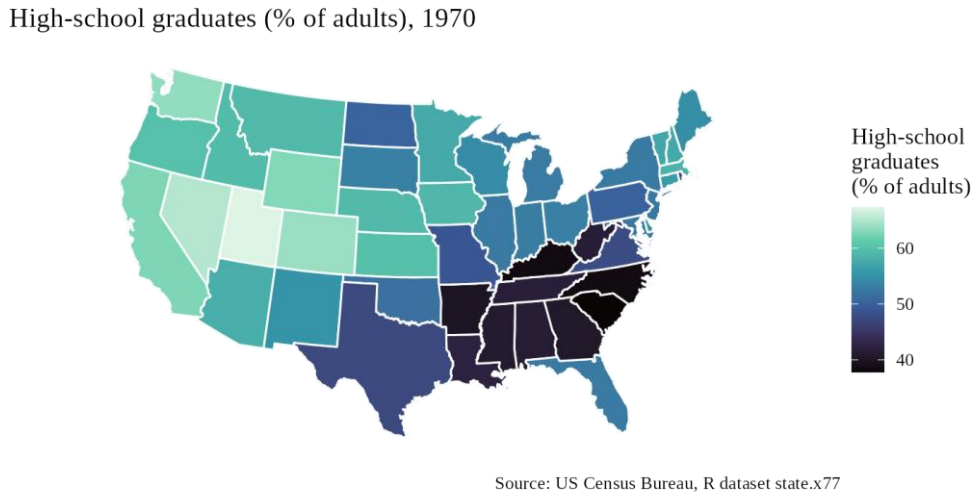


Figure 1: Choropleth map of high-school graduates (% of adults), 1970

Ans: The highest state is Utah with 67.3% and the lowest is South Carolina with 37.8%. The median of all 50 states is 53.25%. (Alaska and Hawaii are used in the statistics but are not shown on the map.)

Task 2: The regional story

Code:

```
reg <- tapply(V, d$zone, mean)
for (z in names(reg)) cat(z, "=", round(reg[z], 2), "\n")
cat("Highest region:", names(which.max(reg)), round(max(reg), 2), "\n")
cat("Lowest region :", names(which.min(reg)), round(min(reg), 2), "\n")
cat("Gap =", round(max(reg) - min(reg), 2), "\n")
cat("cor(V, W) =", round(cor(V, W), 2), "\n")
```

Output:

```
Northeast = 53.97
South = 44.34
North Central = 54.52
West = 62
Highest region: West 62
Lowest region : South 44.34
Gap = 17.66
cor(V, W) = 0.62
```

Ans: The mean of hsgrad is 53.97 in the Northeast, 44.34 in the South, 54.52 in the North Central and 62.00 in the West. The West is the highest and the South is the lowest, and the gap between them is 17.66 percentage points.

The correlation between hsgrad and income is r = 0.62, which is moderate and positive: states with more high-school graduates tend to have a higher per-capita income, but this is not proof that one causes the other.

Task 3: Classes change the story

Code:

```
d$eq <- cut(V, 5)
br <- quantile(V, seq(0, 1, length.out = 6))
d$qt <- cut(V, br, include.lowest = TRUE)
mapdat$eq <- d$eq[match(mapdat$region, d$id)]
mapdat$qt <- d$qt[match(mapdat$region, d$id)]
cat("Equal interval classes:\n")
for (l in levels(d$eq)) cat("  ", l, ":", sum(d$eq == l), "states\n")
cat("Quantile classes:\n")
for (l in levels(d$qt)) cat("  ", l, ":", sum(d$qt == l), "states\n")
cat("Top class, equal interval: lower limit", round(min(V) + 4 * diff(range(V)) / 5, 1),
    ",", sum(d$eq == levels(d$eq)[5]), "states\n")
cat("Top class, quantile: lower limit", round(br[5], 2),
    ",", sum(d$qt == levels(d$qt)[5]), "states\n")
p3a <- ggplot(mapdat, aes(long, lat, group = group, fill = eq)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "% graduates") +
  labs(title = "Equal interval classes (5)",
       caption = "Source: US Census Bureau, R dataset state.x77") +
  theme_void(base_family = "Times New Roman", base_size = 12)
p3b <- ggplot(mapdat, aes(long, lat, group = group, fill = qt)) +
  geom_polygon(colour = "white") +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "% graduates") +
  labs(title = "Quantile classes (5)",
       caption = "Source: US Census Bureau, R dataset state.x77") +
  theme_void(base_family = "Times New Roman", base_size = 12)
print(p3a)
print(p3b)
```

Output:

```
Equal interval classes:
  (37.8,43.7] : 10 states
  (43.7,49.6] : 4 states
  (49.6,55.5] : 16 states
  (55.5,61.4] : 12 states
  (61.4,67.3] : 8 states
Quantile classes:
  [37.8,45.6] : 10 states
  (45.6,52.6] : 11 states
  (52.6,55.5] : 9 states
  (55.5,59.6] : 10 states
  (59.6,67.3] : 10 states
Top class, equal interval: lower limit 61.4 , 8 states
Top class, quantile: lower limit 59.58 , 10 states
```

Maps:

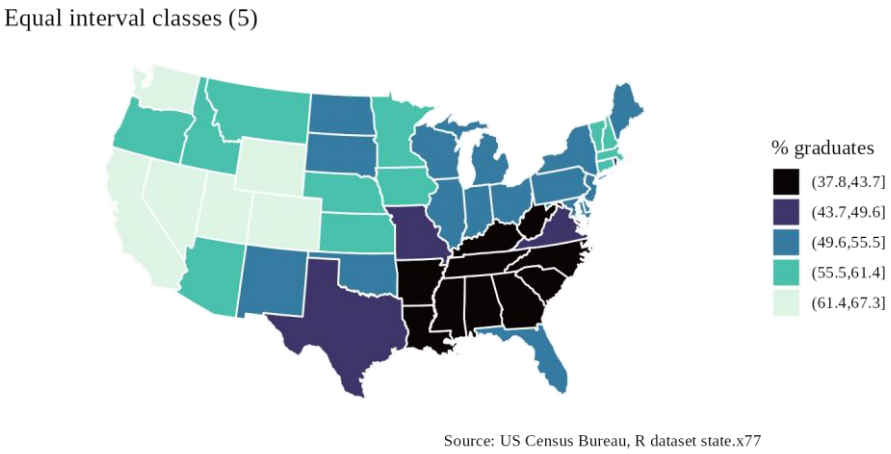


Figure 2: Equal interval classes, cut(V, 5)

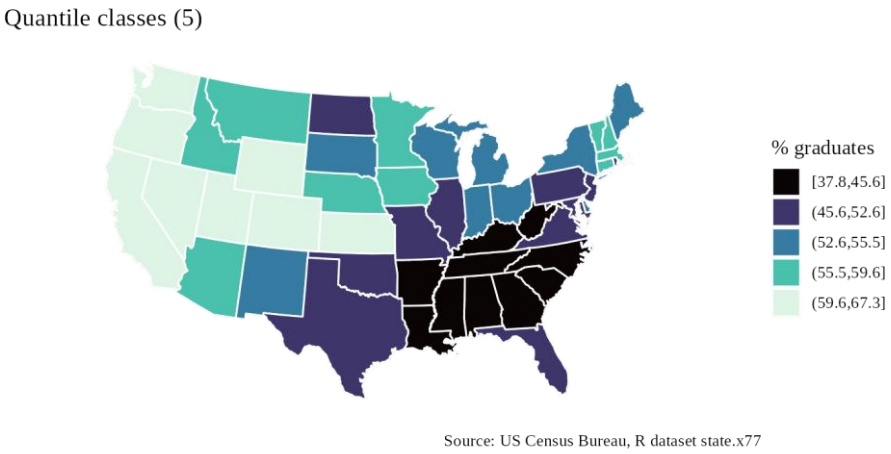


Figure 3: Quantile classes, cut(V, quantile(V, seq(0, 1, length.out = 6)), include.lowest = TRUE)

Ans: In the equal interval map the top class starts at a lower limit of 61.4 and has 8 states. In the quantile map the top class starts at 59.58 (shown as 59.6) and has 10 states. Equal interval makes the highest values look rarer, because only 8 states reach the top colour and 16 states are crowded in the middle class.

I would choose the quantile method for my audience, because every colour has almost the same number of states (9 to 11), so the map shows the differences between states clearly and is easy to read on a poster, even though the class widths are not equal.

Task 4: Counts versus rates

Code:

```
count <- d$pop * 1000 * d$hsgrad / 100
top_c <- order(-count)[1:3]
top_r <- order(-V)[1:3]
cat("Top 3 by count:\n")
for (i in top_c) cat("  ", d$state[i], format(round(count[i]), big.mark = ","), "\n")
cat("Top 3 by hsgrad:\n")
for (i in top_r) cat("  ", d$state[i], V[i], "\n")
cat("States in both lists =", length(intersect(d$state[top_c], d$state[top_r])), "\n")
```

Output:

```
Top 3 by count:
  California 13,269,948
  New York 9,526,052
  Pennsylvania 5,953,720
Top 3 by hsgrad:
  Utah 67.3
  Alaska 66.7
  Nevada 65.2
States in both lists = 0
```

Ans: The top 3 states by count are California, New York and Pennsylvania. The top 3 states by hsgrad are Utah, Alaska and Nevada. No state (0) appears in both lists.

A map of counts would mislead because the biggest states, like California and New York, would always look highest only because they have more people, not because a larger share of their adults finished high school. We should map the rate (hsgrad, % of adults) instead.

Task 5: The story for the South region

Code:

```
S <- d$zone == "South"
cat("Mean hsgrad in South =", round(mean(V[S]), 2), "\n")
cat("Mean hsgrad, all 50  =", round(mean(V), 2), "\n")
cat("Ratio South / all    =", round(mean(V[S]) / mean(V), 2), "\n")
top_s <- which(S)[which.max(V[S])]
cat("Highest state in South:", d$state[top_s], V[top_s], "\n")
cat("South states above national median (", median(V), "):", sum(V[S] > median(V)),
    "out of", sum(S), "\n")
mapdat$south <- mapdat$zone == "South"
p5 <- ggplot(mapdat, aes(long, lat, group = group, fill = qt)) +
  geom_polygon(colour = "white") +
  geom_polygon(data = mapdat[mapdat$south, ], fill = NA, colour = "red", linewidth = 0.6) +
  coord_map("albers", lat0 = 29.5, lat1 = 45.5) +
  scale_fill_viridis_d(option = "mako", name = "% graduates") +
  labs(title = "The South trails the nation in high-school graduation rates",
       subtitle = "High-school graduates (% of adults), 1970; South outlined in red",
       caption = "Source: US Census Bureau, R dataset state.x77") +
  theme_void(base_family = "Times New Roman", base_size = 12)
print(p5)
```

Output:

Mean hsgrad in South = 44.34
Mean hsgrad, all 50 = 53.11
Ratio South / all = 0.83
Highest state in South: Delaware 54.6
South states above national median (53.25): 1 out of 16

Map:

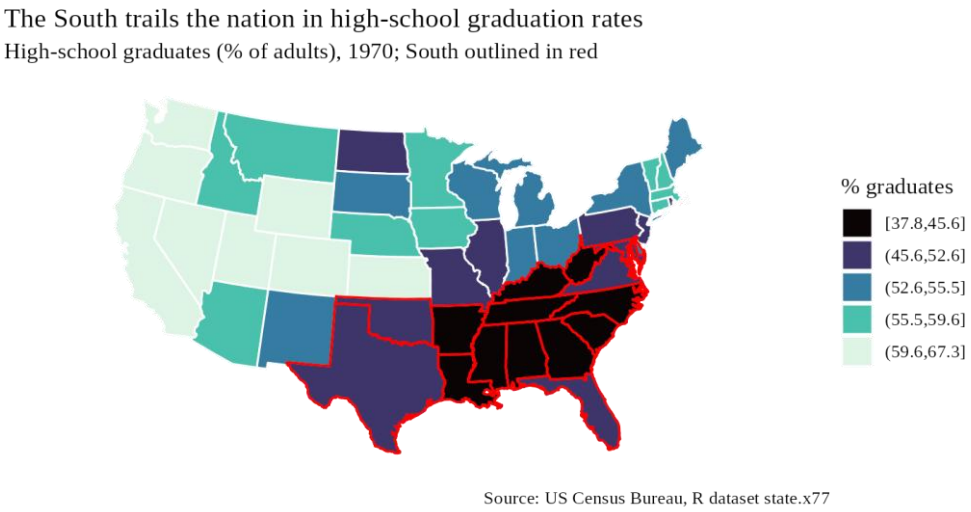


Figure 4: Story map with the South outlined in red

Ans: The mean of hsgrad in the South is 44.34%, and its ratio to the mean of all 50 states (53.11%) is 0.83. The highest state in the South is Delaware with 54.6%. Only 1 South state out of 16 is above the national median.

Headline: The South Trails the Nation in High-School Graduation Rates

Evidence 1: The South averages 44.34% high-school graduates, only 0.83 times the national average of 53.11%.

Evidence 2: Just 1 of the 16 Southern states, Delaware (54.6%), is above the national median of 53.25%, while South Carolina is the lowest in the country at 37.8%.

Recommendation: For the classroom poster, use the quantile map of rates (not counts) with the South outlined, a clear legend and the source note.